Vector databases

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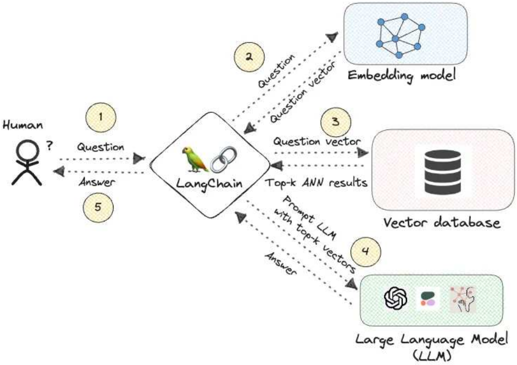
Retrieval Augmented Generation (RAG)

- Provides LLM accurate and relevant information for generating the response.
- Leverages external data source.
- Complementary strategy to LLM fine tuning how to improve relevance of the LLM's answers.



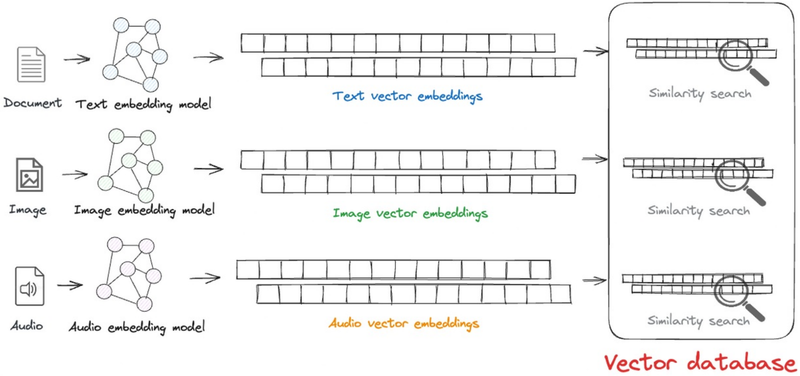
Source: `https://thedataquarry.com/blog/vector-db-2/`

Vector database

- Designed to store (high-dimensional) vectors.
- Effective search capabilities for similar vectors.
- Allow to store also vector metadata and/or original object from which the vector was derived.

Vector embedding

- Input data (text, audio, video) is transformed to high-dimensional vector by embedding model/transformer.
- Embedding captures some properties (usually semantics meaning) of the input data.
- Allows to do a similarity search across set of the vectors.



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Vector embedding

- Similar embeddings correspond to objects which are also **semantically** similar.
- Vector embeddings have other interesting properties, e.g. well know

$$king - man + woman \approx queen \tag{1}$$

Metric types

To be able to do similarity search, we have to define how to measure similarity - we have to define metric on given vector space.

- Metric measure the distance of two vectors.
- It express similarity among the vectors.
- Usually smaller distance means bigger similarity, but there are also metrics where bigger number means bigger similarity.

Metric examples

- L2 (Euclidian)

$$d(x,y) = \sqrt{\sum_{i=1}^{n}(x_i - y_i)^2}$$

- L1 (Manhattan)

$$d(x,y) = \sum_{i=1}^{n}|x_i - y_i|$$

- scalar/inner product (vector need to be normalized)

$$x \cdot y = \sum_{i=1}^{n} x_i \cdot y_i$$

- cosine similarity

$$\cos\Theta = \frac{x \cdot y}{|x||y|}$$

- Jaccard index (similarity of two sample sets)

$$J(A,B) = \frac{|A \cap B|}{|A| + |B| - |A \cap B|}$$

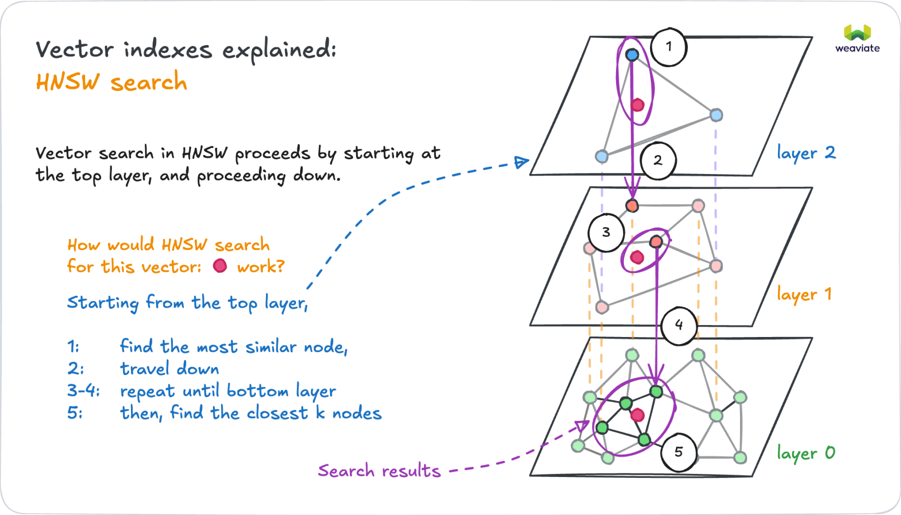
Vector index

- Enabled effective **similarity** search in high-dimensional vector space.
- As in relational DB, updating index adds overhead during data insertion.
- Some index types sacrifice precision for speed.

Vector indexes examples

- Inverted file (IVF)
  - Splits data into several clusters using e.g. K-means clustering.
  - Query identifies the most similar cluster and search only this cluster.
  - Has many variants like IVFFlat, IVFPQ (IVF product quantization), . . .
- Locality-sensitive hashing (LSH)
  - Hash-based algorithm, vectors are clustered based on the fuzzy hashing function.
- Hierarchical Navigable Small World (HNSW)
  - Graph-based algorithm, nearest neighbours are connected.
  - Index organized in several layers, higher level are more sparse.

HNSW search



Source: `https://docs.weaviate.io/weaviate/concepts/vector-index`

Popular vector database

- All major relational databases now support vector data types and related functions.
- Major search engine supporting vectors

Elasticsearch Opensearch

- Specialized vector database

Weavite Pinecone

Chroma Milvus Qdrant

- And many more ...

Vector Database Demo - pgvector

```
docker run --rm --name postgres -e POSTGRES\_PASSWORD=postgres -d -it -p 5432:5432
PGPASSWORD=postgres psql -h localhost -U postgres postgres
```

```
CREATE EXTENSION IF NOT EXISTS vector;
CREATE TABLE vect_test (id serial PRIMARY KEY, embedding vector(3));
INSERT INTO vect_test(embedding) VALUES ('[1,1,1]'), ('[2,2,2]'),
('[3,3,3]'), ('[4,4,4]'), ('[5,5,5]'), ('[5,5,6]'), ('[5,6,5]');
SELECT * FROM vect_test ORDER BY embedding <-> '[5,5,5]' limit 3;
SELECT * FROM vect_test WHERE embedding <->  '[5,5,5]' < 2;
SELECT * FROM vect_test WHERE embedding <+>  '[5,5,5]' < 2;
SELECT * FROM vect_test WHERE embedding <=>  '[5,5,5]' = 0;
```

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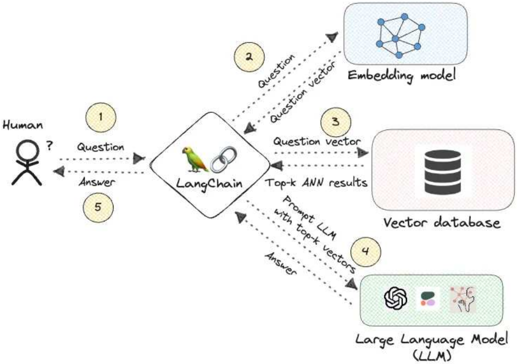

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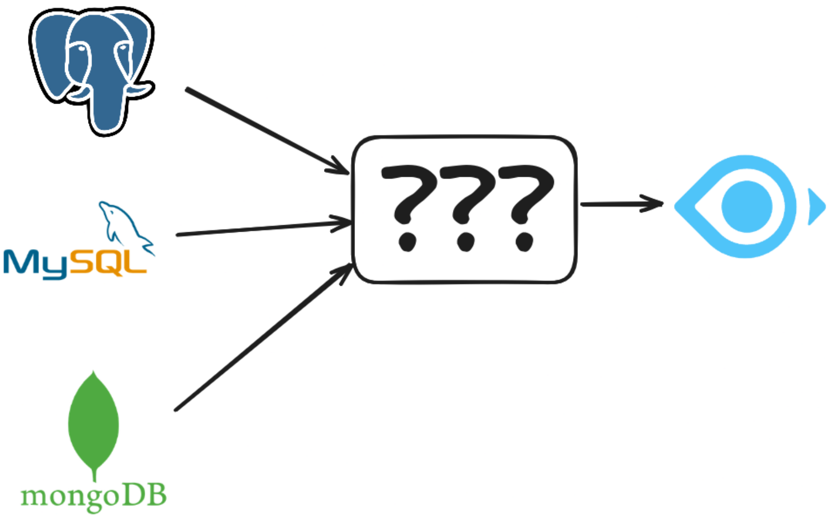
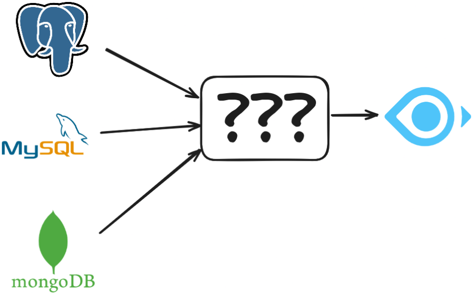
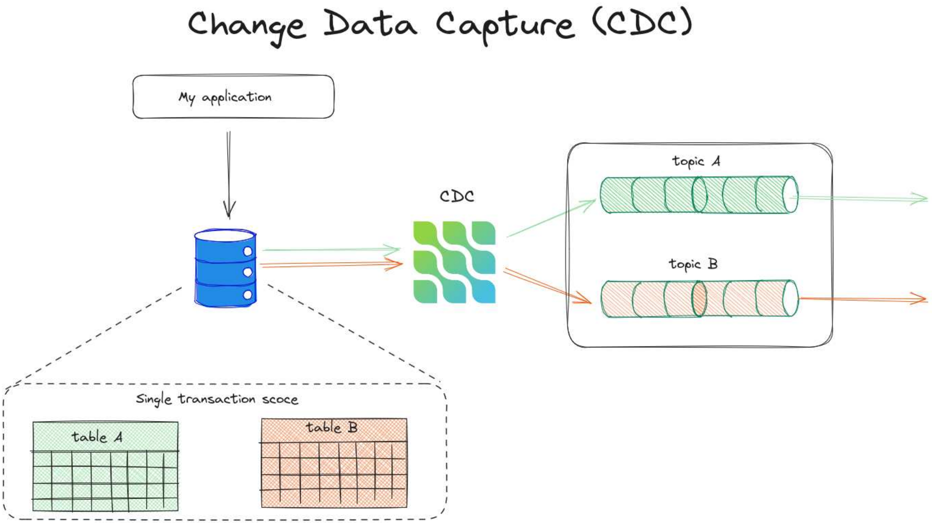

MySQL
mongoDB
???



- Consistent data, no data losses, no dual writes.
- Get all the changes without any delay in the real-time.
- Not overload the DB with the queries.


Change Data Capture (CDC)
My application
CDC
topic A
topic B
Single transaction scoce
table A
table B

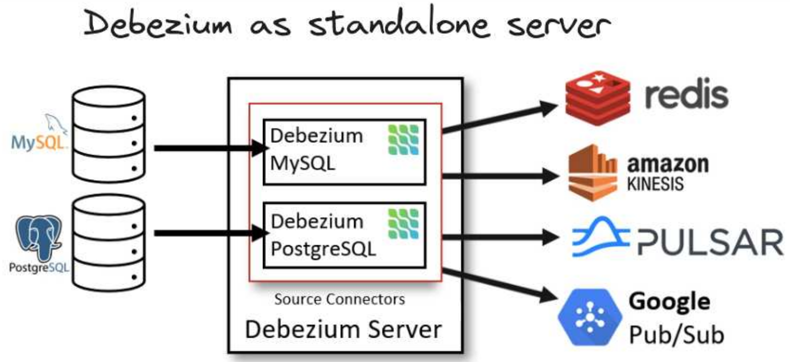
Debezium

debezium

- Leading CDC framework, de-facto industry standard.
- Fully open source: `https://github.com/debezium/`
- Supports all major databases, including non-relational databases.
- Integrations with many 3rd-party tools and frameworks.
- Large and active user community.
- Used by many companies in production (see Debezium public references).

Debezium can be deployed as

- Kafka source connector
- standalone server
- library which can be embedded into Java application



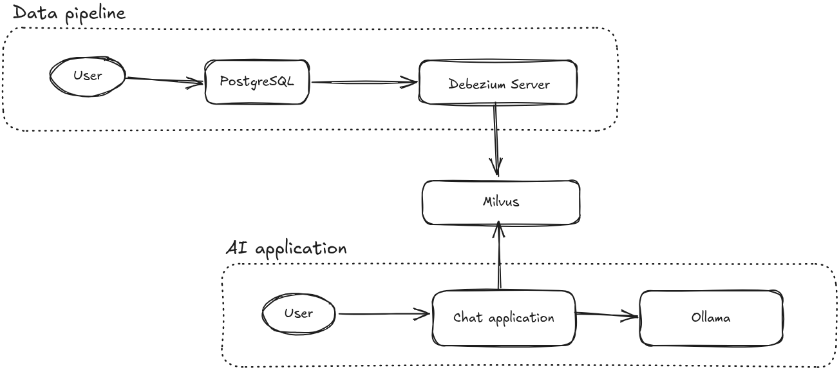
RAG with Debezium

- Debezium provides AI module which allows to create embeddings as a single message transform (SMT).
- Integrated with major LLM provides.
- Currently Debezium server provides direct integration with Milvus, Qdrant and InstructLab.
- For Kafka Connect we provide JDBC sink connector, also dedicated Kafka Connect sink connectors for Milvus and others exists.
- No need to do any changes to the legacy applications or adjusting data flows.
- Modernization of the app stack can happen in parallel to existing applications without any distraction.

RAG Demo: PostgreSQL, Debezium, Milvus

See

- https://github.com/debezium/debezium-examples/tree/main/ai-rag
- https://debezium.io/blog/2025/05/19/debezium-as-part-of-your-ai-solut



Source: https://debezium.io/blog/2025/05/19/debezium-as-part-of-your-ai-solution/

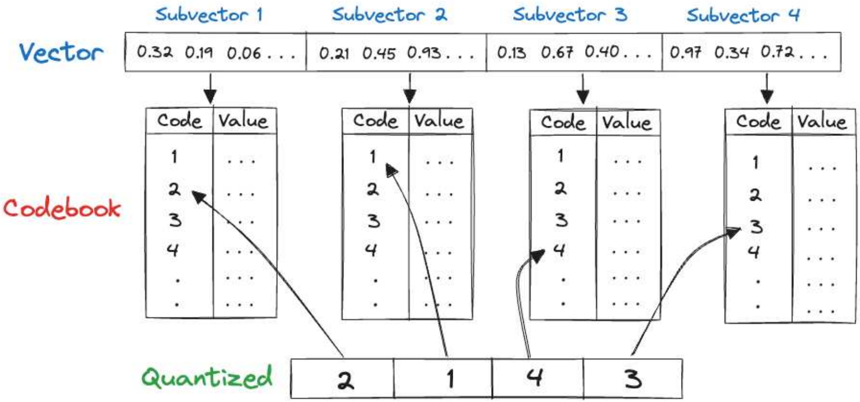
Questions?

Thank you!

Questions?

Backup slides

Product quantization



Source: `https://lancedb.github.io/lancedb/concepts/index_ivfpq/`